IIMT3642: Managing and Mining Big Data - Classification (SVM)

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Outline

- Classification and Linear Models
- Support Vector Machine
  - Margin Maximization
  - Soft Margin
  - Kernel

Reading for today's lecture:
Data Science for Business, Foster Provost and Tom Fawcett, O'Reilly, 2013.
pp. 92 - 94

Classification and Linear Models

- Classification is the process of learning a model (from training data) that can assign an instance of interest (whose class is unknown) to one of the predefined classes.
- The classification process consists of two steps:
  - Step 1: Learn a classification model from training data;
  - Step 2: Apply the model to classify an instance with unknown class.

- A linear model employs a linear combination of descriptive attributes, namely decision boundary or *f(x)*, to classify instances.

$$f(x) = w_0 + w_1 x_1 + w_2 x_2 + \ldots + w_n x_n$$

In which, values of weights (*w*) are to be learned from data

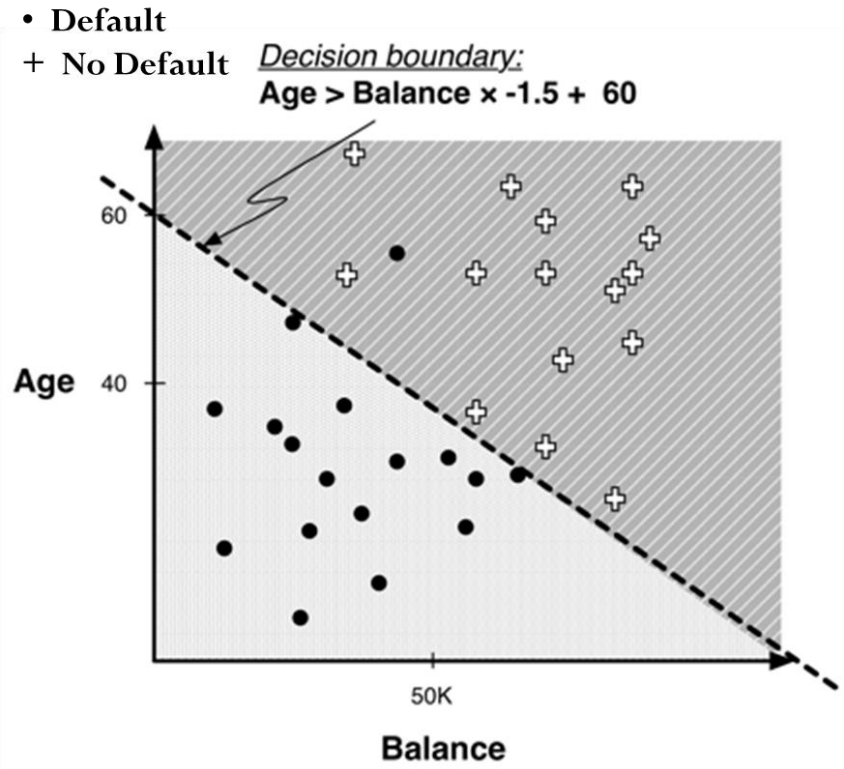
- A new instance (not in the training data) will be classified as one class if *f(x) > 0* or the other class if *f(x) ≤ 0*

f(x) as a Decision Boundary:

$$w_0 + w_1 \times Age + w_2 \times Balance = 0$$

e.g.,

$$-60 + Age + 1.5 \times Balance = 0$$

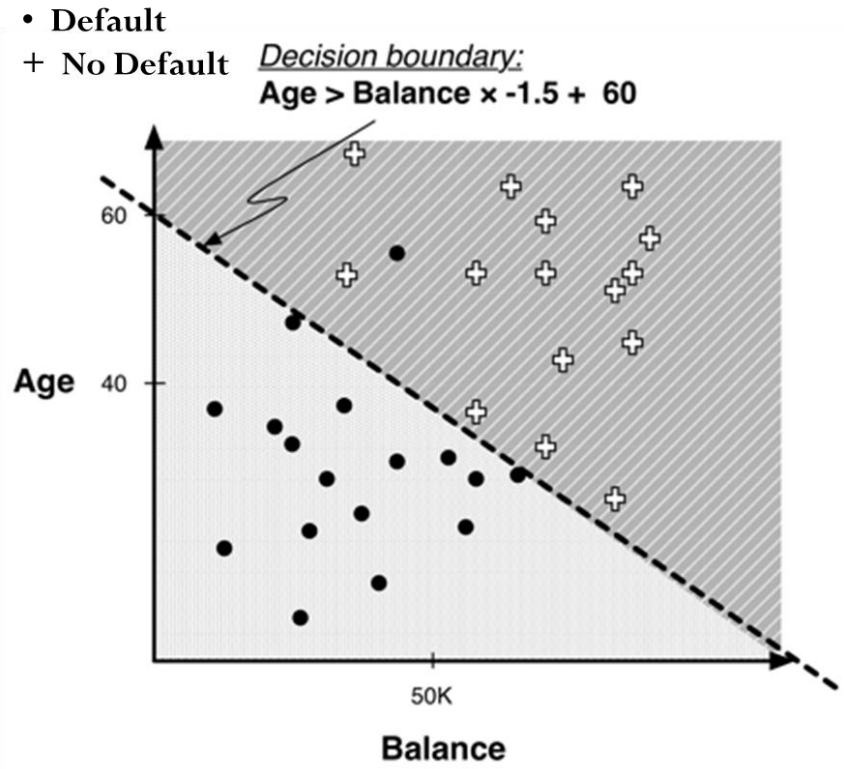


Classification using the decision boundary:

Label the incoming test instance as

$\oplus$ if $-60 + Age + 1.5 \times Balance > 0$

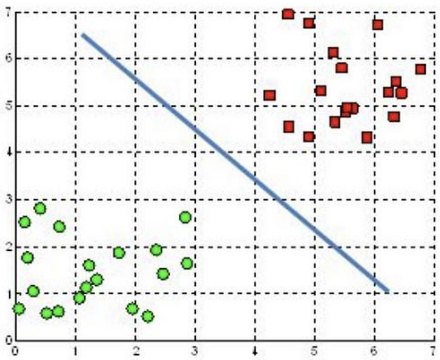
$\bullet$ if $-60 + Age + 1.5 \times Balance \leq 0$



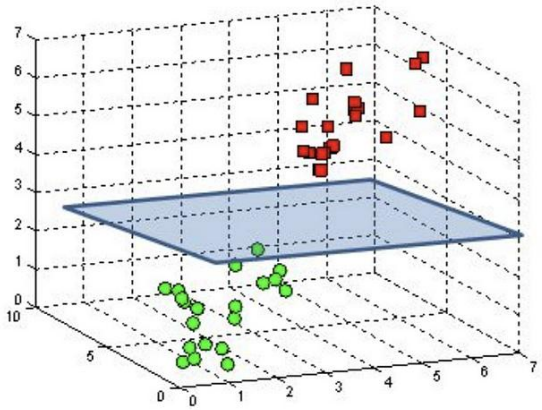
Maximum Margin Hyperplane

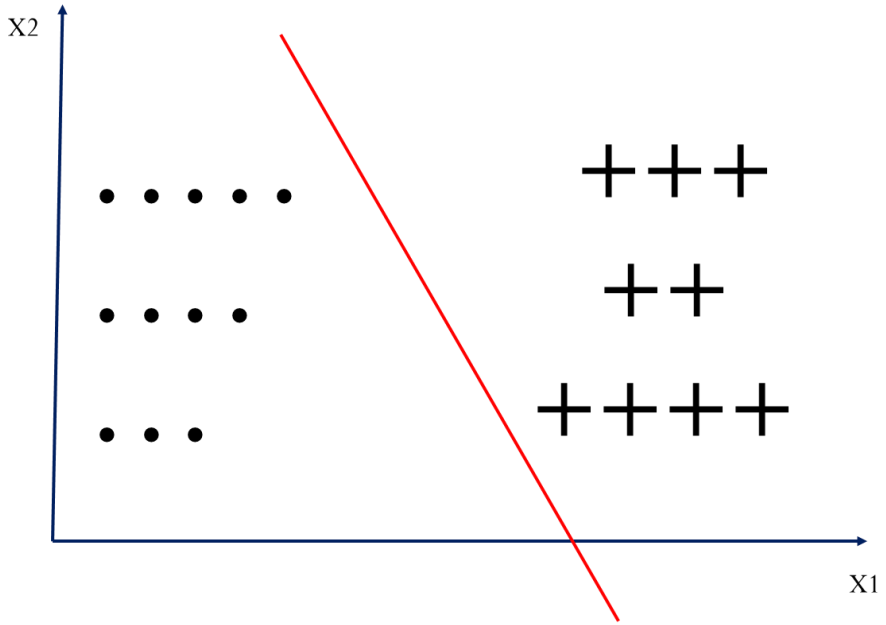
- In multi-dimensional space, decision boundary is a hyperplane that separates one class from the other:
  - if there are only two descriptive attributes: a hyperplane reduces to a line
  - use two-attributes data to illustrate in the next

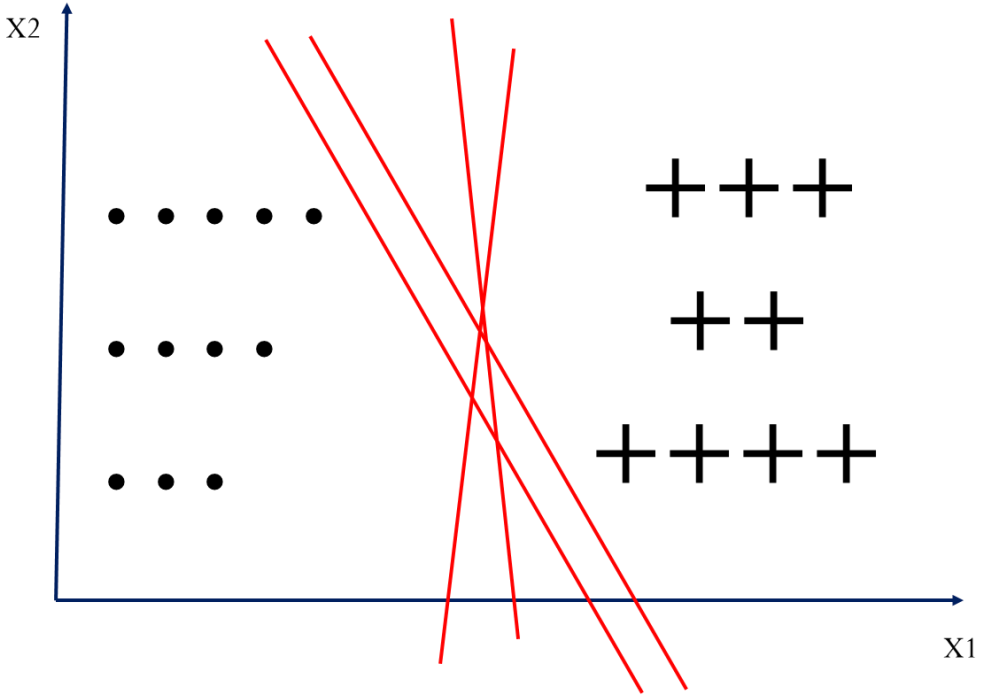
A hyperplane in $\mathbb{R}^2$ is a line



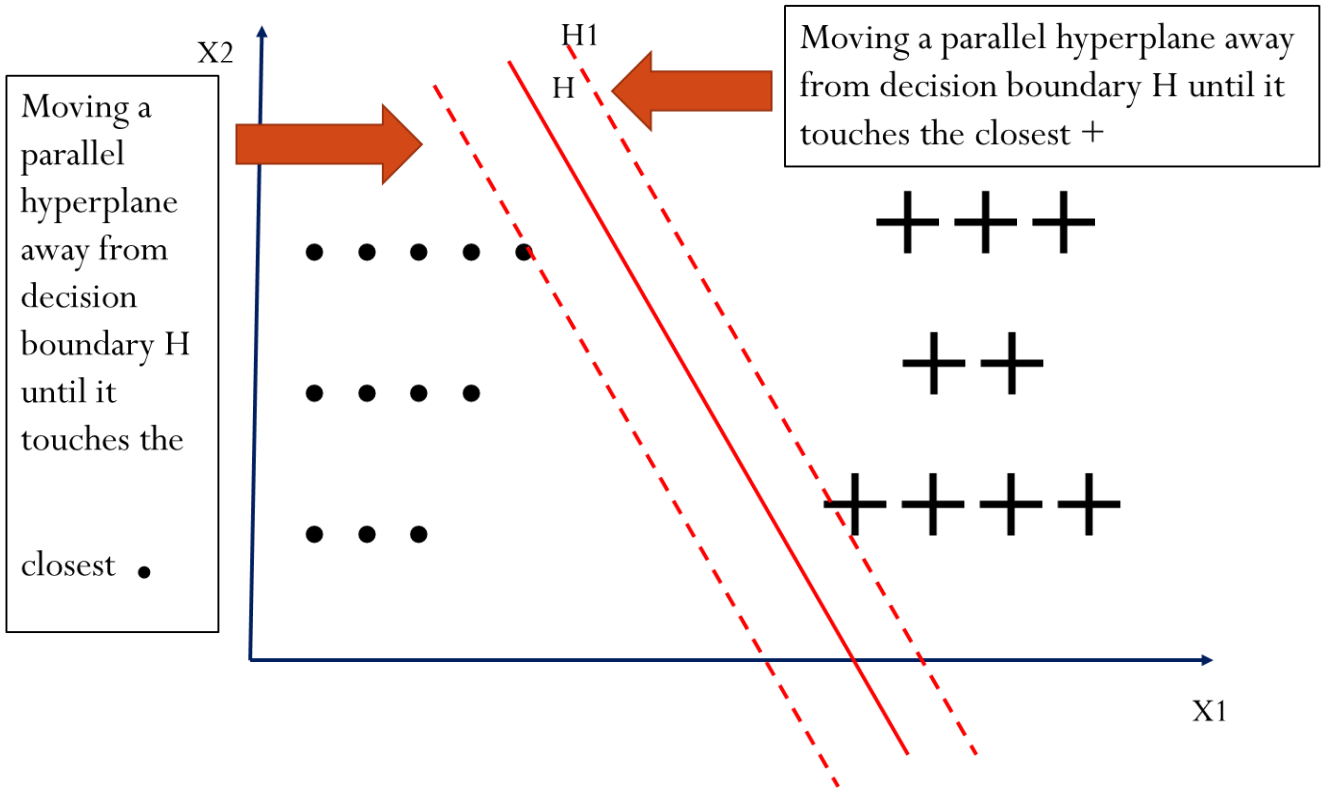
A hyperplane in $\mathbb{R}^3$ is a plane

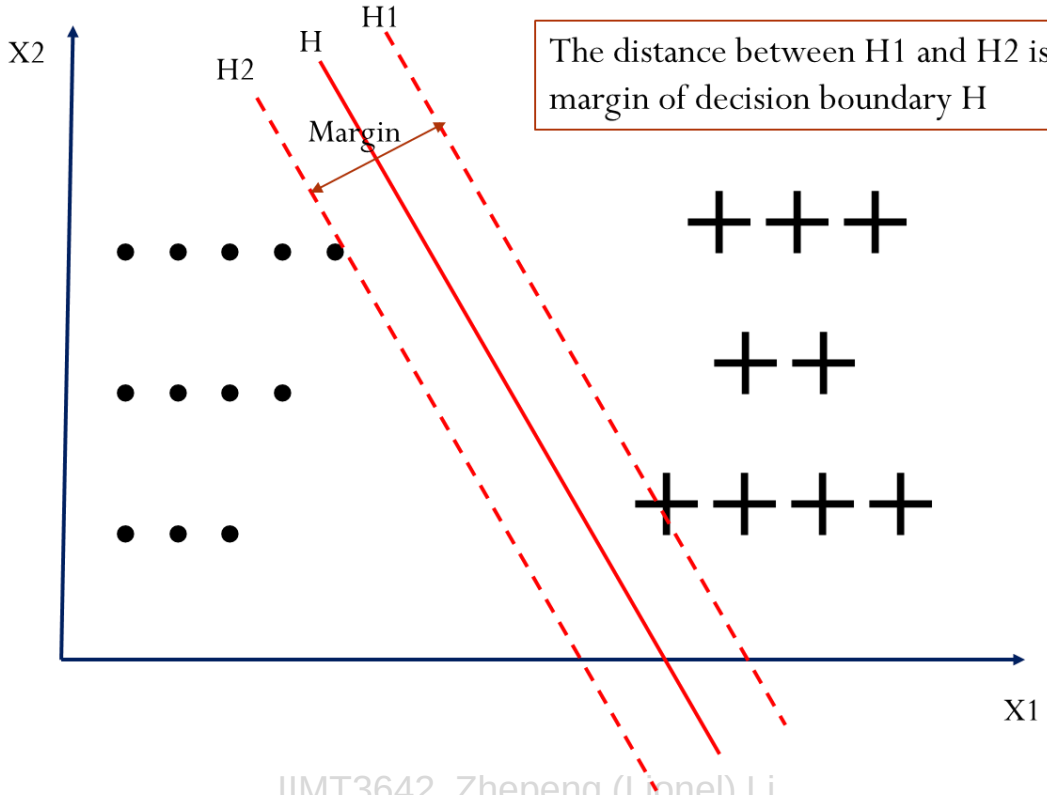



X2
X1


X2
X1

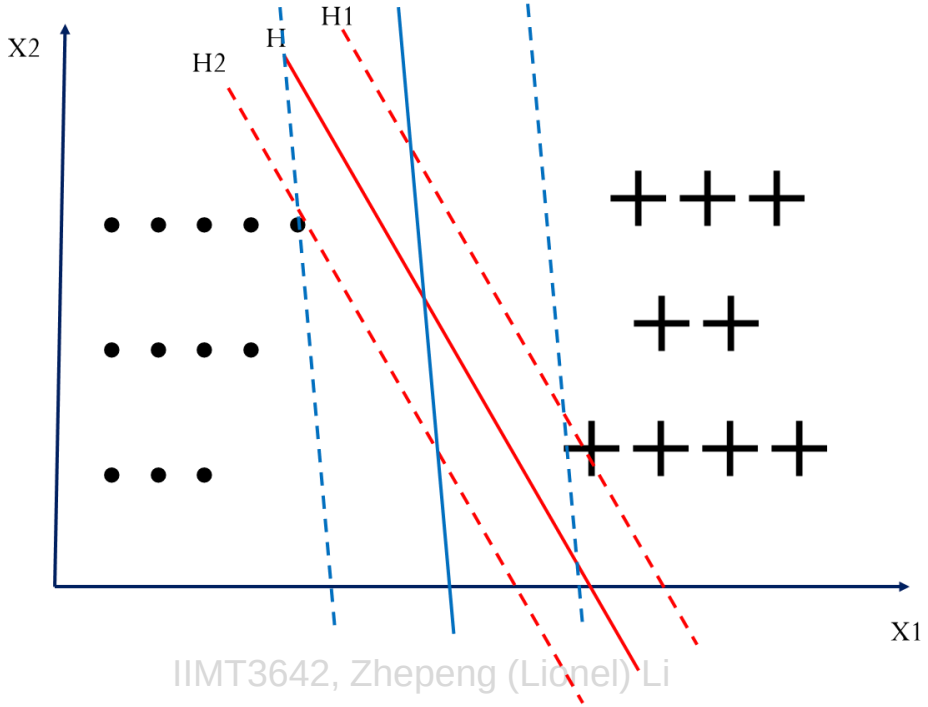
- There are infinite number of hyperplanes; each of them perfectly classifies training examples, i.e., training error = 0
- The best hyperplane should perform the best in classifying test (unseen) examples, i.e., minimizing test error.
- How to choose the best hyperplane: using K-fold cross validation?
- But there are too many hyperplanes to choose from


Moving a parallel hyperplane away from decision boundary H until it touches the closest •
X2
H1
H
Moving a parallel hyperplane away from decision boundary H until it touches the closest +
X1

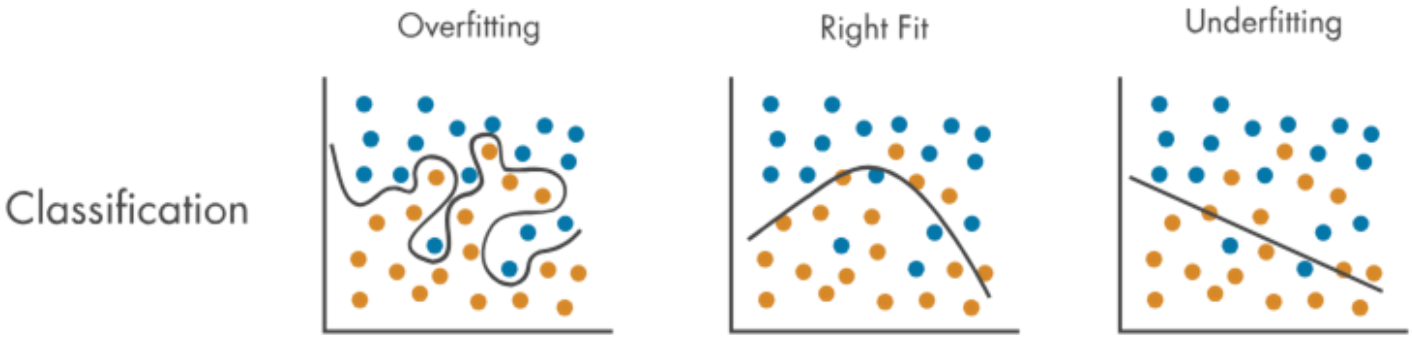


The distance between H1 and H2 is the margin of decision boundary H

Question: Which decision boundary incurs smaller test error, red or blue?



- Blue decision boundary (hyperplane) incurs smaller test error
  - Intuitively, blue hyperplane has larger margin, thus more rooms for unseen test examples
  - Theoretically, smaller margin hyperplane tends to overfit training data and thus red hyperplane incurs larger test error



Support Vector Machine

- Support vector machine (SVM) finds the best decision boundary by searching for the hyperplane with the largest margin.

- SVM becomes an Optimization problem:
  - Maximize margin

    Subject to (constraints)

    all training examples above H1 belong to the “+” class

    all training examples below H2 belong to the “·” class

- Mathematically (for your knowledge):

$$H : w_0 + w_1x_1 + w_2x_2 + \ldots + w_nx_n = 0$$
$$H1 : w_0 + w_1x_1 + w_2x_2 + \ldots + w_nx_n = 1$$
$$H2 : w_0 + w_1x_1 + w_2x_2 + \ldots + w_nx_n = -1$$
$$Margin = \frac{2}{\sqrt[2]{w_1^2 + w_2^2 + \ldots + w_n^2}}$$

- Mathematically (for your knowledge):
  - For a training example $< x_1, x_2, \ldots, x_n, y >$
    - Let *y=1* if the example belongs to the class plus
    - Let *y=-1* if the example belongs to the class dot
  - Such that a good classifier should have

$$w_0 + w_1 x_1 + w_2 x_2 + \ldots + w_n x_n \geq 1 \ if \ y = 1$$
$$w_0 + w_1 x_1 + w_2 x_2 + \ldots + w_n x_n \leq -1 \ if \ y = -1$$

- Mathematically (for your knowledge):
  - The SVM optimization problem is formulated as below.

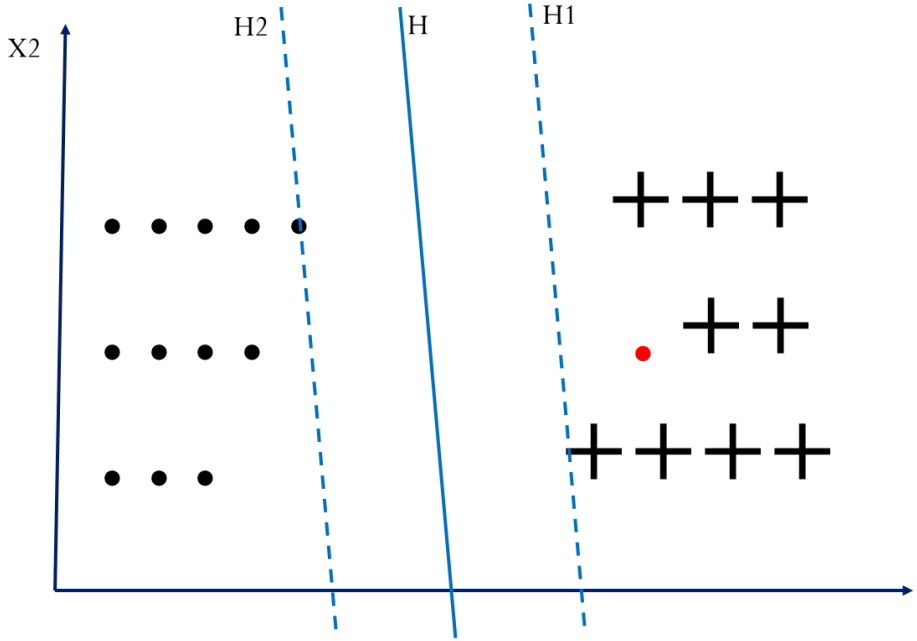
$$\text{Maximize } Margin = \frac{2}{\sqrt{w_1^2 + w_2^2 + \ldots + w_n^2}}$$

$$\text{Subject to: for each training example}$$

$$w_0 + w_1 x_1 + w_2 x_2 + \ldots + w_n x_n \geq 1 \text{ if } y = 1$$

$$w_0 + w_1 x_1 + w_2 x_2 + \ldots + w_n x_n \leq -1 \text{ if } y = -1$$

- Misclassification error



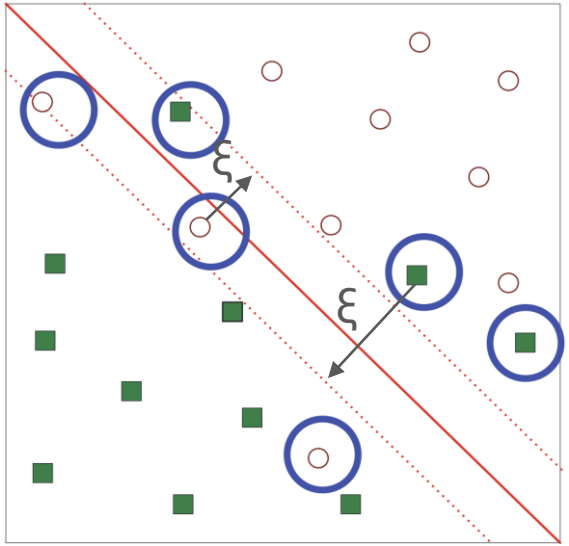
- Real-world data can be complicated: linearly non-separable
  - define a penalty for each misclassified training example (e.g., red dot in the prior slide)
  - Considering non-separable cases, we can introduce slack variables to incorporate misclassification penalties as a cost. So, the optimization problem is updated to soft margin:

Let $\xi_i = \max[0, 1 - y_i \cdot (\vec{\omega} \cdot \vec{x}_i + \omega_0)]$

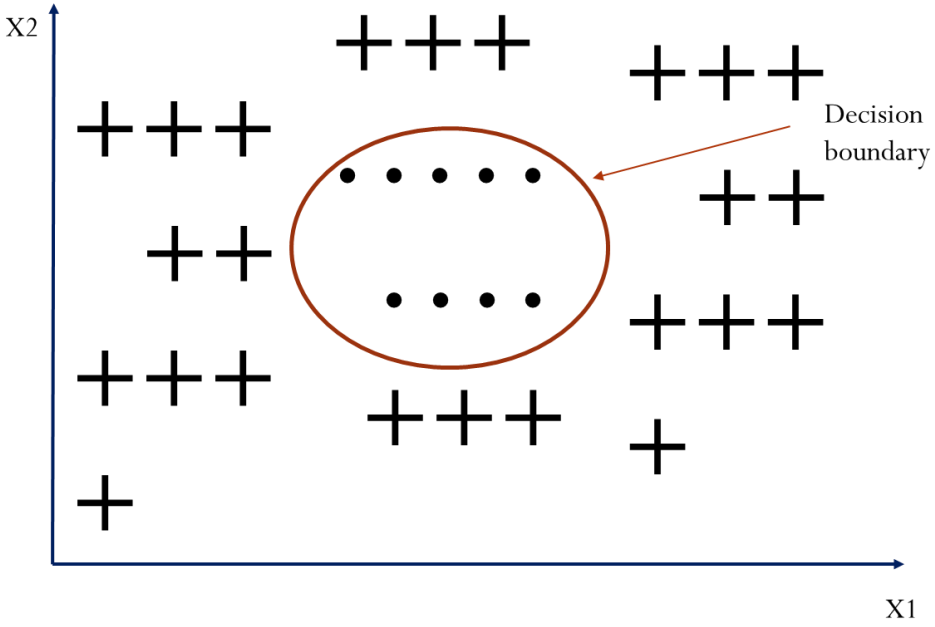
$$\text{Minimize } L(\omega) = \frac{\|\vec{\omega}\|^2}{2} + C \times \left(\sum_{i=1}^{N} \xi_i \right)$$

Subject to:

$$y_i = \begin{cases} 1 \text{ If } \vec{\omega} \cdot \vec{x}_i + \omega_0 \geq 1 - \xi_i \\ -1 \text{ If } \vec{\omega} \cdot \vec{x}_i + \omega_0 \leq -1 + \xi_i \end{cases}$$



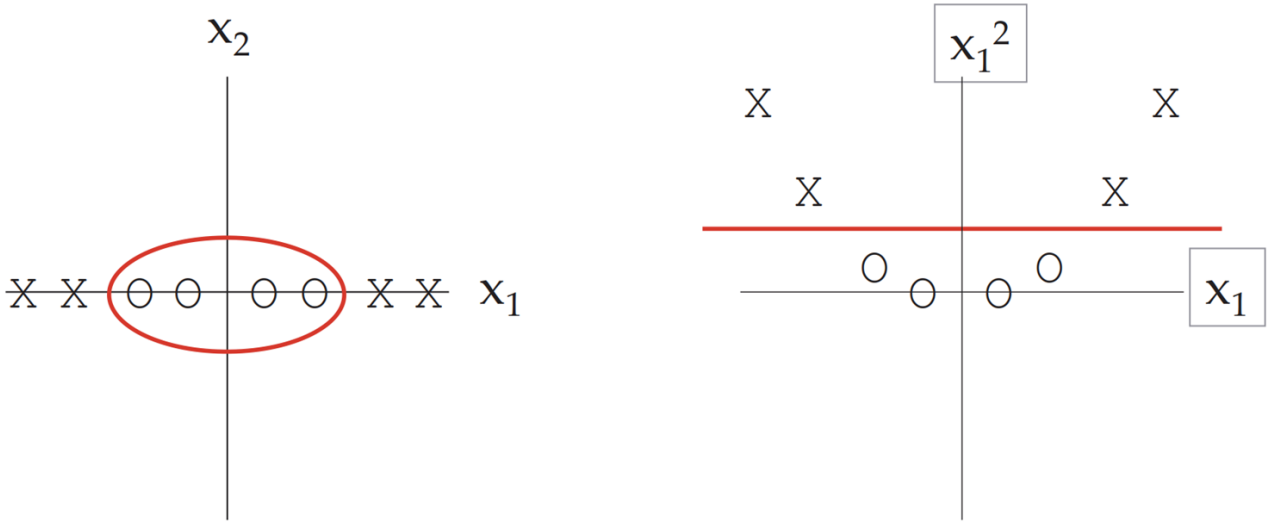
Non-linear decision boundary



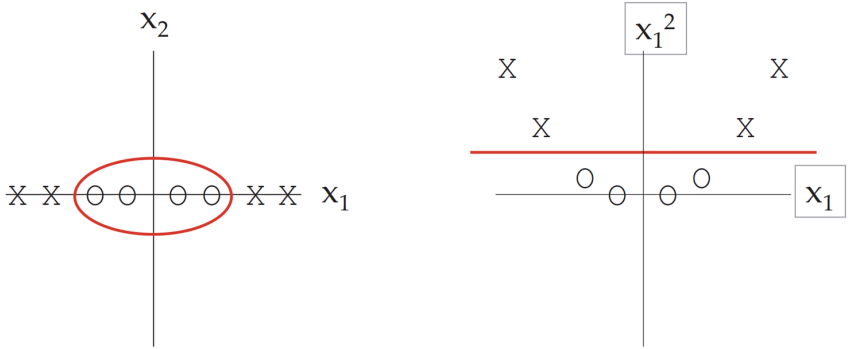
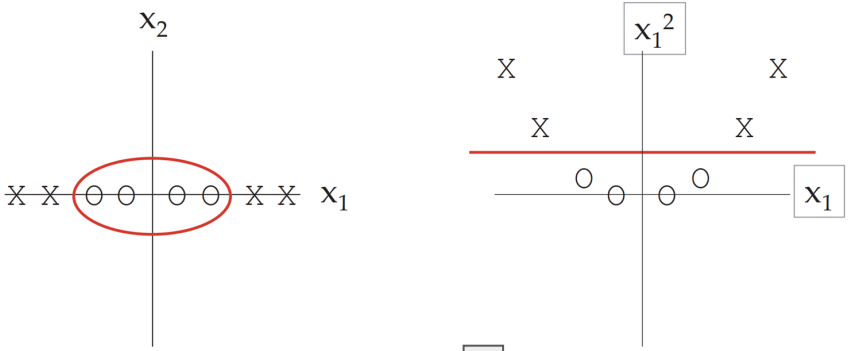
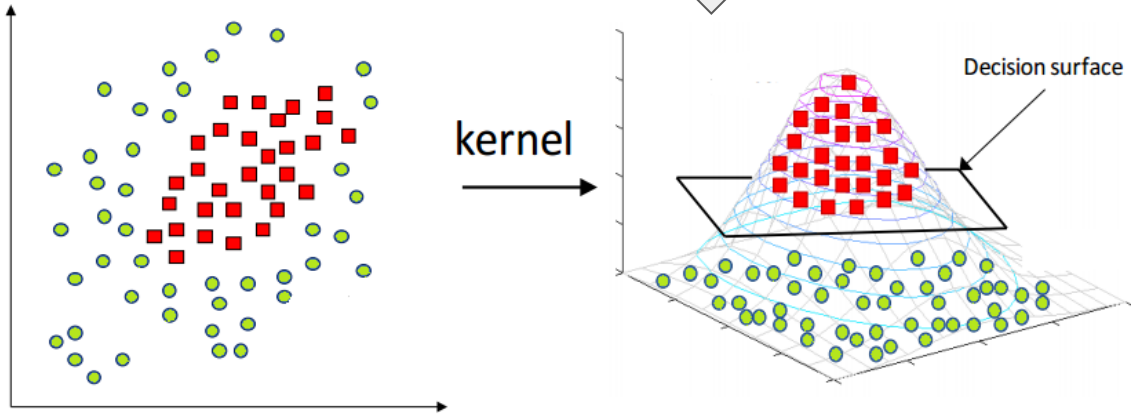
- Real-world data can be complicated: non-linear decision boundary
  - There is no straight line which can separate one class from the other or decision boundary is not a line
  - It should not be classified by a linear model
- The nonlinearity issue can be overcomed by
  - Apply Kernel function to transform original training examples into the case that can be separated by a line
  - Then apply Linear Support Vector Machine discussed in previous slides

Apply Kernel function to transform original training data

- A simple example:



- Kernel function is $\phi : x \to X = \phi(x),\ e.g.,\ \phi(x_1, x_2) = \left(x_1, x_2, x_1^2, x_2^2, x_1 x_2\right)$
- Rather then running SVM on x, run it on $\phi(x)$
  - Finding non-linear separator in input space is equivalent to using non-linear kernels!



- The underlying idea is that
  - Data points are more separatable when mapping to a higher dimensional space
- Common Kernels to choose:
  - RBF
  - Gaussian
  - Polynomial
  - Linear